PG No. 8 D_{2h} mmm [orthorhombic]

Table 1: Representation matrices

Irrep.						
A_g	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{010}:\begin{bmatrix}1\end{bmatrix}$	$2_{100}:\begin{bmatrix}1\end{bmatrix}$	$-1:\begin{bmatrix}1\end{bmatrix}$	$m_{001}:\begin{bmatrix}1\end{bmatrix}$
	$m_{010}:\begin{bmatrix}1\end{bmatrix}$	$m_{100}:\begin{bmatrix}1\end{bmatrix}$				
B_{1g}	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{010}:\begin{bmatrix}-1\end{bmatrix}$	$2_{100}:\begin{bmatrix}-1\end{bmatrix}$	$-1:\begin{bmatrix}1\end{bmatrix}$	$m_{001}:\begin{bmatrix}1\end{bmatrix}$
	$m_{010}:\begin{bmatrix}-1\end{bmatrix}$	$m_{100}:\begin{bmatrix}-1\end{bmatrix}$				
B_{2g}	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$2_{010}:\begin{bmatrix}1\end{bmatrix}$	$2_{100}:\begin{bmatrix}-1\end{bmatrix}$	$-1:\begin{bmatrix}1\end{bmatrix}$	$m_{001}:\begin{bmatrix}-1\end{bmatrix}$
	$m_{010}:\begin{bmatrix}1\end{bmatrix}$	$m_{100}:\begin{bmatrix}-1\end{bmatrix}$				
B_{3g}	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$2_{010}:\begin{bmatrix}-1\end{bmatrix}$	$2_{100}:\begin{bmatrix}1\end{bmatrix}$	$-1:\begin{bmatrix}1\end{bmatrix}$	$m_{001}:\begin{bmatrix}-1\end{bmatrix}$
	$m_{010}:\begin{bmatrix}-1\end{bmatrix}$	$m_{100}:\begin{bmatrix}1\end{bmatrix}$				
A_u	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{010}:\begin{bmatrix}1\end{bmatrix}$	$2_{100}:\begin{bmatrix}1\end{bmatrix}$	$-1:\begin{bmatrix}-1\end{bmatrix}$	$m_{001}:\begin{bmatrix}-1\end{bmatrix}$
	$m_{010}:\begin{bmatrix}-1\end{bmatrix}$	$m_{100}:\begin{bmatrix}-1\end{bmatrix}$				
B_{1u}	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{010}:\begin{bmatrix}-1\end{bmatrix}$	$2_{100}:\begin{bmatrix}-1\end{bmatrix}$	$-1:\begin{bmatrix}-1\end{bmatrix}$	$m_{001}:\begin{bmatrix}-1\end{bmatrix}$
	$m_{010}:\begin{bmatrix}1\end{bmatrix}$	$m_{100}:\begin{bmatrix}1\end{bmatrix}$				
B_{2u}	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$2_{010}:\begin{bmatrix}1\end{bmatrix}$	$2_{100}:\begin{bmatrix}-1\end{bmatrix}$	$-1:\begin{bmatrix}-1\end{bmatrix}$	$m_{001}:\begin{bmatrix}1\end{bmatrix}$
	$m_{010}:\begin{bmatrix}-1\end{bmatrix}$	$m_{100}:\begin{bmatrix}1\end{bmatrix}$				
B_{3u}	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$2_{010}:\begin{bmatrix}-1\end{bmatrix}$	$2_{100}:\begin{bmatrix}1\end{bmatrix}$	$-1:\begin{bmatrix}-1\end{bmatrix}$	$m_{001}:\begin{bmatrix}1\end{bmatrix}$
	$m_{010}:\begin{bmatrix}1\end{bmatrix}$	$m_{100}:\begin{bmatrix}-1\end{bmatrix}$				